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9 August 1965
SUPERSEDING
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9 July 1956
(See 6.4)

MILITARY SPECIFICATION

POWER SUPPLIES, METALLIC RECTIFIER, NAVAL SHIPBOARD

1. SCOPE

1.1 Scope - This specification covers general requirements for metallic rectifier type power supplies for naval shipboard use.

1.2 Classification - Power supplies shall be of the following classes as specified (see 6.1):

Class R - Regulated

Class NR - Nonregulated

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose)

MILITARY

MIL-C-25 - Capacitors, Fixed, Paper-Dielectric, Direct-Current (Hermetically Sealed in Metallic Cases), General Specification for.
MIL-C-62 - Capacitors, Fixed, Electrolytic (DC, Aluminum, Dry Electrolyte, Polarized) General Specification for.
MIL-S-901 - Shock Tests, H.I. (High Impact), Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-I-1361 - Instruments, Electrical Measuring, Shunts, Resistors and Transformers (Ship and Shore Use).
MIL-R-2033 - Relays for Naval Shipboard Electrical Service.
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-C-2174 - Controllers, Direct-Current, Naval Shipboard.
MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard.
MIL-C-3965 - Capacitors, Fixed, Non-Solid Electrolytic, (Tantalum, Foil and Sintered-Slug), General Specification for.
MIL-M-10304 - Meters, Electrical Indicating, Panel Type, Ruggedized, General Specification for.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-F-15160 - Fuses, Instrument, Power, and Telephone.
MIL-S-15291 - Switches, Rotary, Snap Action.
MIL-L-15719 - Lubricating Grease (High-Temperature, Electric Motor, Ball and Roller Bearings).

FSC 6130

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MIL-M-16034 - Meters, Electrical-Indicating (Switchboard and Portable Types).
MIL-O-16485 - Ohmmeters Insulation-Resistance-Indicating, Portable.
MIL-T-16784 - Terminal Boards.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.
MIL-S-18396 - Switches, Meter and Control, Naval Shipboard.
MIL-R-19523 - Relays, Auxiliary, Naval Shipboard.
MIL-C-19978 - Capacitors, Fixed, Plastic (or Paper-Plastic) Dielectric (Hermetically Sealed in Metallic, Ceramic, or Glass Cases), General Specification for.
MIL-F-21346 - Fuseholders, Block and Plug Type and Associated Electrical Clips, General Specification for.
MIL-S-21604 - Switches, Rotary, Multipole and Selector Type, 1 to 10 Amperes.
MIL-C-26655 - Capacitors, Fixed, Solid Electrolyte, Tantalum, General Specification for.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-701 - Preferred and Guidance Lists of Semiconductor Devices.
MIL-STD-740 - Noise Measurements of Shipboard Machinery and Equipment.
MIL-STD-761 - Electric Power, Alternating Current for Shipboard Use, Characteristics and Utilization of.

DRAWINGS

BUREAU OF SHIPS

9000-S6202-F-73907 - Light, Indicator, Switchboard, 2 Lamp-SPF, Types B-27A Through B-27G.
9000-S6202-74228 - Fuse Holder, Double Dead Front Blown Fuse Indicator Type FHL10G, 30 Amp, 120V, 1-1/4 Inch by 1/4 Inch Fuse.
9000-S6202-74229 - Fuse Holder, Single for 1-1/4 Inch by 1/4 Inch Fuses with Blown Fuse Indicator, Type FHL11G.
9000-S6202-74230 - Fuse Holder, Single Dead Front, Blown Fuse Indicator Type FHL12G, 450V, 1-3/32 Inch by 1-1/2 Inch Fuses.

(See supplement 1 for list of applicable detail specifications.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 100016).

3. REQUIREMENTS

3.1 Detail requirements. - Detail requirements shall be as specified in the applicable detail specification. In the event of any conflict between the requirements of this specification and the individual detail specification, the latter shall govern (see 6.1).

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3.2 Qualification. - The power supply furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein and has been listed on or approved for listing on the applicable qualified products list.

3.3 General requirements. - The equipment shall comply with the requirements of MIL-E-917, except as specified herein or in the individual detail specification. Where a requirement of this specification conflicts with a requirement of MIL-E-917, the requirement of this specification shall govern.

3.4 Definitions. - The definitions specified in 3.4.1 through 3.4.7 shall apply to the various technical terms wherever such terms appear in this specification.

3.4.1 Rectifying element. - A rectifying element is a circuit element which conducts current effectively in one direction only.

3.4.2 Power supply. - A power supply is an assembly used to convert alternating current (a.c.) power to direct current (d.c.) power. It does not include rotary machinery such as motor generators or dynamotors. For the purpose of this specification, a power supply shall be considered to be all the components, subassemblies and assemblies necessary to give specified performance.

3.4.2.1 Power supply, regulated (class R.). - A regulated power supply is one which makes use of special components having the primary function of automatically controlling output voltage, current, or both.

3.4.2.2 Power supply, nonregulated (class NR). - A nonregulated power supply is one which does not make use of any components having a primary function of automatically controlling output voltage, current, or both. Output regulation is entirely dependent on the inherent regulation of basic components of power transformers and power rectifying elements and the function of the dummy loads, if used. Class NR power supplies may contain magnetic amplifiers or silicon controlled rectifiers for adjustment of output voltage.

3.4.3 Battery charger. - A battery charger is a device consisting of one or more components utilizing an alternating current source to reactivate batteries in which reversible chemical reaction occurs. For purpose of this specification, a battery charger, using metallic rectifiers for converting a.c. power to d.c. power, is also a power supply.

3.4.4 Full-wave rectification. - Full-wave rectification is rectification in which both halves of the a.c. cycle are transmitted as unidirectional current.

3.4.5 Ripple voltage. - The ripple voltage is the alternating component of a substantially unidirectional voltage.

3.4.6 Percent ripple voltage. - The percent ripple voltage is the ratio in percent of the effective (r.m.s) value of the ripple voltage in the pulsating voltage to the average value of the pulsating voltage.

3.4.7 Dummy load. - A dummy load is a component part of a power supply (usually a resistor) which is connected across the output circuit for the purpose of partially loading the power supply to improve output performance (usually voltage stability or regulation or both) over the rated load range. When resistive, it is frequently termed a bleeder resistor.

3.5 Design requirements. -

3.5.1 Operation. - Power supplies shall provide d.c. output as required by the individual detail specification. All power supplies shall have a transformer input so that the d.c. output circuit is isolated from the a.c. input. Rectification and control (that is, regulation) of any power handled by the equipment shall be accomplished by static components such as magnetic amplifiers, metallic rectifiers or silicon controlled rectifiers. Power-handling rectifier circuits shall provide full-wave rectification.

3.5.2 Electrical rating and performance. -

3.5.2.1 A.c. voltage. - Equipment shall be designed to operate from Type I power as described by MIL-STD-761. Input voltage (normally either 440 or 115 volts) shall be as specified in the individual

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detail specification (see 6.1). Equipment shall operate as specified without adjustment with variations of plus or minus 5 percent (average line-to-line, 3 phase) of rated voltage. Likewise, performance shall not be affected by 3 percent unbalance between phase voltages.

3.5.2.1.1 A.c. input transients. - When the a.c. input supply voltage suffers a momentary (total duration up to 2.0 seconds) rise or drop of up to 18 percent of the nominal value the power supply shall:

(a) Be affected in neither calibration or adjustment, nor shall the life of any component be shortened.

(b) Suffer a change of output voltage of a value not to exceed that of the a.c. input, percentage-wise. For example if the a.c. voltage changes by 10 percent, the d.c. output voltage shall not change by more than 10 percent.

3.5.2.2 Frequency. - Unless otherwise specified in the individual detail specification (see 6.1) equipment shall be designed to operate on a supply voltage with frequency of 60 cycles per second (cps). Equipment shall operate as specified (see 6.1), without adjustment, with variations of plus or minus 5 percent of rated frequency.

3.5.2.3 Phases and balances. - Equipment shall be designed for operation on a single phase or 3-phase, 3-wire ungrounded power source as specified in the individual detail specification. In case of a 3-phase, 3-wire ungrounded circuit, current consumed by the equipment shall be balanced to the extent that the current in any wire shall not differ by more than 10 percent from the average current in the three wires.

3.5.2.4 D.c. output. - All equipment shall operate over the entire output range as specified (see 6.1) with variation of voltage, frequency and ambient as specified in 3.5.2.1, 3.5.2.2 and 3.5.3.2. This shall include any special output performance requirements such as voltage regulation and adjustment, current regulation and adjustment, automatic voltage cutoff or transfer, current limiting, current overload, transient performance and ripple voltage content. By definition, class R power supplies shall always have automatic control of output voltage, current or both.

3.5.2.4.1 Remote sensing-Where remote sensing is required for operation, or is permitted or specified by an individual detail specification, the power supply shall be fitted with terminals which may be connected at installation to a remote load point so that the power supply can compensate (or compound) for voltage drop between its output terminals and the load terminals. Current drawn by such a sensing circuit shall not exceed 1 ampere. (See 3.5.2.8 for required fail-safe features associated with remote sensing circuits).

3.5.2.5 Duty. - Unless otherwise specified in the individual detail specification, equipment shall be designed to operate continuously at the specified rating. If intermittent duty is specified, a clear description of the duty cycle shall be given.

3.5.2.6 Type of load. - The type of load may be battery charging, capacitive, inductive (including d.c. motor), resistive or any combination which may be specified in the individual detail specification. Unless otherwise specified, battery charging loads shall be considered to be lead-acid type batteries. Regardless of the type of load specified, equipment shall operate satisfactorily at rated output voltage and current when connected to any of the above types of load, without hunting or oscillation of output voltage. For purposes of this specification, oscillation shall be defined as any sustained variation of output voltage due to interaction between power supply and load, for example, when the load is a regulated motor-generator.

3.5.2.7 Voltage ripple. - The maximum allowable voltage ripple in the output shall normally be specified in terms of percent. Unless otherwise specified in the individual detail specification the meaning of percent value shall be as defined herein (see 3.4.6). The wave form of the a.c. component of the output voltage shall be such that the peak-to-peak value of voltage ripple shall not exceed three times the r.m.s. value when both quantities are measured in accordance with 4.5.10.6.

3.5.2.8 Circuit and component protection means. - The following safety and fail-safe design features shall be provided in all power supplies:

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(a) Circuits using adjustable devices shall be arranged so that no damage to any component can be caused by excessive current or voltage due to any possible setting of the adjustable device. For example where a series resistor is adjusted to control current through a circuit element either sufficient rating of all circuit elements, or a fixed current-limiting resistor shall be used to prevent damage to circuit elements if the adjustable resistor is set at minimum value.

(b) Power supplies using voltage sensing leads for remote voltage regulation shall be so circuited that voltage regulation at the terminals of the power supply is automatically provided if the sensing lead circuit is broken.

(c) Power supplies shall be provided with fuses for overload and short circuit on output terminals.

(d) Power supplies shall be provided with fuses to protect surviving semi-conductor rectifiers from damage when one rectifier fails by becoming a short circuit.

(e) Filter capacitor circuits shall be fused to protect the power supply in the event of a short-circuited capacitor. Capacitors for filtering of power circuits shall be connected in at least 2 independently fused circuits.

3.5.2.9 Efficiency.- Design of equipment shall be such as to give the maximum practical efficiency without deviation from other requirements herein and in the contract or order. When dummy loads are used, the power dissipated therein shall not exceed 1 percent of normal full load rating.

3.5.3 Environmental requirements.-

3.5.3.1 Mechanical.-

(a) Shock.- Equipment shall meet the HI shockproof requirements for grade A, deck mounted, Class I as determined by Tape A test of MIL-S-901.

(b) Vibration.- Equipment shall meet the vibration requirements of type I of MIL-STD-167.

(c) Inclined operation.- Equipment shall give no change in operation when inclined at an angle of 45 degrees from the vertical in any direction.

3.5.3.2 Ambient temperature.- Unless otherwise specified in the contract or order and in the individual detail specification, the rating of equipment shall be based upon an ambient temperature range of 23° to 50°C. The specified ambient (that is, 50°C., 40°C. and other temperatures is the maximum ambient rating and shall always include the range of ambient down to 23°C.

3.5.3.3 Relative humidity.- Equipment whether or not energized, shall withstand 100 percent relative humidity for a period of 168 hours at room temperature plus or minus 10°C. During this period the value of insulation resistance between all transformer and reactor windings and ground and between insulated windings of the same transformer or reactor shall be not lower than the "minimum value curve" of figure 1. The value of dissipation factor for the transformers and reactors shall not exceed the "maximum value curve" of figure 2. Following humidification there shall be no evidence of material damage or excessive corrosion of any components or hardware. The electrical performance shall not be affected by humidification.

3.5.4 Airborne and structureborne noise.-

3.5.4.1 Airborne noise.- Airborne noise generated by the power supply under any operating conditions shall not exceed the limits of grade B in MIL-STD-740.

3.5.4.2 Structureborne noise.- When specified (see 6.1), structureborne noise shall not exceed the limits of type 3 equipment of MIL-STD-740.

3.5.4.3 Use of resilient mounting of power supply components.- The preferred method of equipment noise reduction shall be by design and construction of circuits and components used therein. Where components are mounted in the power supply assembly by resilient mounting means (sound isolators), exposed metal parts of such components (for example, transformer cores) shall be connected to the equipment frame by a grounding lead, to reduce potential electric shock hazard.

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3.5.5 Electromagnetic (radio frequency) interference.- Design of power supply, its circuits and components shall be such as to produce minimum conducted and radiated interference within the frequency range of 150 kilocycles to 30 megacycles. No specific limits for this interference apply unless specified by an individual detail specification or contract or order (see 6.1).

3.6 Construction requirements.-

3.6.1 Enclosure.- Equipment shall be furnished with enclosures in accordance with MIL-E-2036. Unless otherwise specified in the individual detail specification (see 6.1), enclosure type shall be dripproof protected. For purposes of this specification, a "dripproof protected enclosure" shall be defined as one which excludes drop of liquid or solid particles striking the enclosure at an angle of 45 degrees from the downward verticle, and also excludes entrance of a 1/2 inch diameter rod through louvers or ventilation grills. The "degree of exclusion of drops of liquid or solid particles" and the "definition of failure to meet these requirements" as specified by MIL-STD-108 shall apply. Enclosures and framework shall be constructed of steel, aluminum or both, and shall be of the dead-front type. Enclosures, unless otherwise specified in the individual detail specification (see 6.1), shall have a front mounted hinged access door. A latch to hold the door open despite motion of the ship in which the equipment is installed shall be provided.

3.6.2 Mounting.- Unless otherwise specified in the individual detail specification (see 6.1), equipment weighing 300 pounds or less shall be designed for bulkhead mounting. Equipment weighing in excess of 300 pounds shall be designed for deck or platform mounting. All mounting provisions shall be arranged for easy access and adjustment of mounting screws or bolts without use of special tools and without disassembly of equipment. Portable units shall be equipped with carrying handles. Unless otherwise specified (see 6.1) portable equipment shall have provisions for both deck mounting and bulkhead mounting and shall be tested as bulkhead mounted equipment for all requirements specified herein. All equipment with provision for deck mounting shall have mounting feet or runners of sufficient elevation to provide easy installation and to prevent blockage of any bottom ventilation openings from accumulation of dust.

3.6.3 Lifting attachments.- Equipment weighing 150 pounds and over shall be provided with eye-bolts or equivalent means for lifting and handling. The lifting attachments shall be so located that when lifting vertically the equipment does not tilt more than 5 degrees from the vertical.

3.6.4 Size and weight.- Maximum size and weight shall be as specified in the individual detail specification. Where size and weight limits are not specified, the equipment shall be designed for the minimum values consistent with operation and service required. Measurement of size shall cover maximum dimensions of height, width, and depth. Dimensions shall exclude extensions due to lifting eye-bolts but shall include extension due to mounting supports, dripshields, ventilation louvres and protruding components such as meters, indicator lights, switch handles and similar items. Measurements of weights shall include all material (assemblies and subassemblies) of the equipment.

3.6.4.1 Submarine equipment.- Cohen required and specified (see 6.1) in the individual detail specification or contract or order, equipment shall be packaged to pass through 25 inch diameter submarine hatch and watertight door. (A watertight door is 36 inches high, 20 inches wide and has 10 inch radius corners (the largest clear rectangular portion of the opening is 18 inches high and 20 inches wide)). A minimum degree of disassembly will be permitted for this access.

3.6.5 Ventilation.- Ventilation shall be either natural draft or forced draft as specified in the individual detail specification. In either case the dimensions of the equipment shall include any necessary allowance for air intake and discharge. If method of ventilation is not specified method shall be natural draft.

3.6.5.1 Natural draft.- Equipment with natural draft cooling shall be designed to give maximum cooling of components with minimum size and weight. Unless otherwise restricted by contract or order or approved by the bureau or agency concerned, this shall be accomplished by construction of enclosures and arrangement of components to give a chimney effect natural draft with height of equipment the maximum dimension.

3.6.5.2 Forced draft - The following requirements shall apply:

(a) Fan and fan motor shall comply with commercial practice except that electrical insulation and corrosion prevention treatment shall comply with MIL-E-917.

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(b) Fan and motor bearings shall be sealed types requiring no lubrication. Preferred types of bearing are type 120 or type III class 3 or class 8 of FF-B-171. Preferred lubricant is silicone grease conforming to MIL-L-15719.

(c) Preferred (not mandatory) direction of forced air flow is from top to bottom of equipment.

(d) Screens and warning plates shall be used to minimize the probability and severity of personnel injury due to contact with moving fan blades.

(e) Fan and motor shall sustain no damage during shock and vibration tests of the assembled equipment.

(f) Forced draft ventilated equipment shall be cooled by a single direct-drive fan.

3.6.5.2.1 Fan failure protection shall be provided to avoid equipment damage from accumulation of heat upon failure of forced draft ventilation devices. Such protection shall operate to automatically deenergize the equipment at the a.c. input. Upon failure of the ventilating system, the automatic protection devices shall deenergize the equipment before any semiconductor devices exceed their maximum allowable operating temperature. The following devices, listed in order of preference, may be used to accomplish the above protection function.

(a) Temperature-sensitive switch, located adjacent to a heat source, the heat from which is normally removed by the properly operating ventilating system. Upon failure of forced draft ventilation, the heat accumulates and actuates the switch.

(b) Temperature-sensitive switch, mounted in intimate contact with a heat sink of a power-handling semiconductor device.

(c) Centrifugally-operated switch associated with fan motor.

(d) Air-flow operated vane switch.

3.6.6 Arrangement of components.- Unless otherwise specified in the individual detail specification components ordinarily used in the operation of the equipment shall be operable without opening any access doors. These components shall typically include on-off switches, meters, indicator lights and voltage adjusting devices. The following components shall be accessible for operation, adjustment, replacement and test through a front hinged door. Access to these components shall not require removal of any other components.

(a) Test points or terminals (see 3.6.6.1)

(b) Adjustable calibration devices.

(c) Switch, relay and contactor contacts.

(d) Encapsulated components

(e) Semiconductor components

(f) Fuses and manual-reset overload protection devices

(g) Terminal blocks used for connection of external wiring

3.6.6.1 Test points.- Test points shall be provided for all control circuits for making measurements of voltage, current, and resistance as necessary for trouble shooting and adjusting operation of the equipment.

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3.7 Material requirements. -

3.7.1 Insulation. -

3.7.1.1 Insulation resistance. - The insulation resistance of all circuits shall be not less than 10 megohms when measured with all circuits cold.

3.7.1.1.1 Design and construction of power supply components shall be such that a 500 volt (high internal impedance ohmmeter) insulation resistance measurement will not damage any components under the following conditions:

(a) Output (and remote sensing if used) terminals connected together.

(b) Ohmmeter connected between those terminals and ground.

(c) No insulation failure exists, that is, insulation resistance is above 10 megohms.

(d) Input terminals (2 for single phase equipment. 3 for three phase equipment) connected together.

(e) Conditions (b) and (c).

3.7.1.2 Dielectric strength. - All equipment shall be so designed and constructed as to withstand for a period of 1 minute a dielectric test voltage to ground as follows:

A.c (r.m.s.) input or d.c. output voltage	Test voltage (r.m.s.)
60 and under	600
61 to 90	900
90 to 600	1000 plus twice rated
Over 600	2000 plus 2-1/2 times rated

3.7.2 Painting. - Parts to be painted shall consist of those not having a corrosion-resisting treatment or not fabricated of corrosion-resisting materials of the types specified in MIL-E-917 and shall include exterior and interior surfaces of the enclosure. Painting shall be in accordance with MIL-E-917 except that only one coat of gray enamel need be applied. Marks or scratches from handling during fabrication and testing shall be repainted by either complete repainting of equipment or by touch-up method.

3.8 Components. -

3.8.1 Semiconductor devices. - Semiconductor devices shall be silicon types and shall be selected from MIL-STD-701. For applications which require semiconductor components of ratings other than those listed in MIL-STD-701 the contractor shall obtain approval from the bureau or agency concerned as required by MIL-E-917. Application of semiconductor devices shall be as specified by MIL-E-917. Transistors shall not be used as series regulating elements in output power circuits.

3.8.2 Transformers, reactors and inductors. - Transformers, reactors and inductors shall conform to MIL-E-917. Transformers shall have electrically isolated windings (primary and secondary). Neither auto-transformers nor open-delta connected transformers shall be used.

3.8.3 Meters and meter accessories. - Unless otherwise specified in the individual detail specification, all meters and meter accessories shall be furnished as an integral part of the equipment. When meters are specified this shall be understood to include any necessary meter accessories. Meters shall be in accordance with MIL-M-16034 or MIL-M-10304 (with external zero adjustor) as specified in the individual detail specification. Meter accessories (shunts, resistors, and similar items) shall be in accordance with MIL-I-1361. Meter range shall be as appropriate for the application to give approximately two-thirds full scale deflection at full load conditions.

3.8.4 Indicator lights. - Each power supply shall be provided with an indicator light conforming to Drawing 9000-S6202-73907 complete with lamps. The indicator light shall be connected to indicate when the power supply is energized. The indicator light shall be mounted on the front exterior of the power supply enclosure.

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3.8.5 Fuses.- Fuses shall be provided for protection as follows:

(a) Failure of a power-handling semiconductor device (for example, shorted silicon rectifier). Surviving semiconductor devices shall be protected.

(b) Failure (shorting) or excess leakage of electrolytic filter capacitors of output circuit. Such filter capacitors shall be connected and fused in at least two parallel circuits.

(c) Overload and short circuit protection of output circuit, unless this protection is already provided by other means such as input overload relays independently or in combination with other protective devices specified in (a) and (b).

(d) Pilot circuit protection shall be in accordance with MIL-C-2212. Fuses shall be types F09, F15 through F22 and F60 through F67, of MIL-F-15160 where such fuses provide adequate protection. Fast-acting current-limiting fuses may be used where their characteristics are required.

3.8.5.1 Fuseholders (with indicator).- Fuseholders with blown fuse indicators in accordance with Drawings 9000-S6202-74228, 9000-S6202-74229 and 9000-S6202-74230 shall be provided for fuses as follows:

(a) Output circuit (if fused) except for battery chargers and for power supplies with wide range of adjustment of output voltage.

(b) Input circuit (if fused).

(c) Fuses used for protection of surviving semiconductor devices from damage caused by failure of one or more other semiconductor devices.

3.8.5.2 Fuseholders (without indicators).- Fuseholders without blown fuse indicators shall be clip types in accordance with the applicable types of MIL-F-21346 or the fuseholders required by 3.8.5.1 but with blanke indicator for fuses as follows:

(a) In applications where the required fuse is not usable in an indicator-type fuseholder.

(b) In filter capacitor circuits.

3.8.6 Resistors and rheostats.- Resistors and rheostats shall be selected in accordance with MIL-E-917.

3.8.7 Capacitors.- Capacitors in accordance with MIL-C-3965 (wet tantalum) and MIL-C-26655 (solid tantalum) are preferred. Capacitors in accordance with MIL-C-19978 (plastic dielectric), MIL-C-25 (paper dielectric) or MIL-C-62 (electrolytic) are acceptable. Where the required capacitance is too large to be covered by MIL-C-62, it is permissible to use special sizes provided construction is similar to MIL-C-62. Capacitors of special sizes shall be subject to approval by the bureau or agency concerned.

3.8.8 Contactors and relays.- Contactors, relays and other contact-making control devices shall be in accordance with MIL-C-2212, MIL-C-2174, MIL-R-19523 or MIL-E-2033, as applicable. The use of other special control devices, not covered by specifications referenced herein, will be subject to specific approval of the bureau or agency concerned and will be considered for use only when it has been determined that the use of components, as specified, is not practicable.

3.8.9 Switches.- Switches shall be in accordance with MIL-S-15291, MIL-S-21604, MIL-C-2212, MIL-S-18396 or as otherwise approved by the bureau or agency concerned for a specific application.

3.8.10 Terminal boards.- Terminal boards shall comply with MIL-T-16784 (with insulation in accordance with MIL-E-917) except where the terminal board is integral with and considered a part of another approved component (for example, contactors specified by MIL-C-2212) or where such specified terminal boards are impractical because of application rating or space configuration.

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3.8.11 Miscellaneous component parts.- To best meet equipment performance requirements, it may occasionally be desirable to make use of component parts not covered herein by referenced specifications. These parts shall be considered miscellaneous parts and may include, but may not be limited to special temperature controls or compensating devices and special switches. The use of miscellaneous parts shall be subject to the approval of the bureau or agency concerned. Requests for such approval shall be accompanied by technical data justifying need for and suitability of such parts. Miscellaneous component parts shall comply with MIL-E-917 insofar as practicable. Approval of the bureau or agency concerned shall be obtained for miscellaneous component parts in each specific application.

3.8.12 Prohibited components.- The following parts may not be used:

(a) Motors (except for fans, where permitted)

(b) Variable transformers (typically "Variacs")

(c) Batteries

(d) Meter relays

(e) Electron tubes or other thermionic devices

3.8.13 Prohibited materials.- Materials indicated as prohibited by MIL-E-917 may not be used.

3.9 Drawings.- Drawings shall be furnished in accordance with 3.9.1 through 3.9.3.

3.9.1 Master drawings.- Master drawings shall comply with class A of MIL-D-963, with content as specified herein. All sheets of master drawing shall be of the same size. Minimum size shall be 17 by 22 inches; maximum size shall be 28 by 40 inches. Manuals (see 3.10) shall contain reduced size copies of master drawing. Therefore scale and lettering size of master drawings shall be large enough so that the drawings can be reduced legibly. Master drawings shall contain the following minimum data:

(a) Manufacturer's name and catalog number or equivalent identification of the equipment.

(b) List of descriptive data of the equipment including:

(1) Applicable specification and exceptions.
(2) Complete rating showing:
  (a) Input voltage, frequency, and number of phases.
  (b) Input KW, KVA, current and power factor at zero, 1/2 and full load.
  (c) Output voltage and current at full load.
  (d) Efficiency at one-half and full load.
  (e) Output adjustment range.
(3) Duty (if intermittent, indicate duty cycle).
(4) Shock classification.
(5) Enclosure classification.
(6) Ventilation.
(7) Type of load.
(8) Ambient temperature range.
(9) Ripple content of output voltage.
(10) Weight of equipment (net).
(11) Notation of the corrosion - resistant treatment of all hardware.
(12) Special features.

(c) List of material, presenting following data in tabular form:

(1) Piece number (for flags of outline drawing).
(2) Name of part (for example, transformer, diode, and other similar parts).
(3) Schematic diagram designation for electrical parts (for example, switch SW-1, and other similar designations).

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(4) Number of each part required per power supply assembly.
(5) Military or Federal Specification number or Bureau of Ships drawing number, if any.
(6) Type, class, grade, size or other classification of any referenced specification (see (5)).
(7) Part number or identification assigned by power supply manufacturer.
(8) Name of manufacturer of part.
(9) Part number or identification assigned by part manufacturer.
(10) Federal stock number, if known.
(11) Rating of part from part manufacturer's catalog (for example, resistor 10 ohm 25 watt; diode - 200 PRV, 750 milliamps, and similar ratings.
(12) Application or in use stress on part as used under normal operating condition of power supply assembly, for example resistor 7.1 watts; diode - 91 PRV, 400 milliamps, and other similar applications.

Note - List of material shall exclude ordinary hardware items such as fasteners, component brackets, and other similar hardware items. Heat sink and enclosure assemblies shall be listed. Hookup wire type shall be indicated.

(d) Outline drawing of equipment showing front, side, top and section views as necessary, to present information as follows.

(1) Overall and mounting dimensions.
(2) Location and size of mounting holes.
(3) Approximate arrangement and location of components. Components shall be flagged by piece number.
(4) Location of center of gravity of equipment.
(5) Identification and information plates.
(6) Clearance required for ventilation and to open access doors.
(7) Cable entrance provisions.
(8) Direction of air flow in forced-draft ventilated equipment.

(e) Transformer and reactor data:

(1) Core data:
a. Lamination form - EI, toroid, etc.
b. Lamination material and gage.
c. Stack and interleave dimensions.
d. Construction - open, tape wound cut core, and other construction.
e. Core box, if any.
f. Ground insulation.
(2) Winding data:
a. Insulation class - class B, H, and so forth.
b. Winding identification-same terminal designation as given in schematic and connection diagrams.
c. Wire size and insulation type for example AWG 10 Type B2, and so forth.
d. Number of turns.
e. Voltage.
f. D.c. resistance at 25°C.
g. Material of insulation used between winding layers and between windings.
(3) Varnish impregnation:
a. Varnish type and specification.
b. Varnish manufacturer and material identification.
c. Impregnation procedure - preheating temperature and time, number of dips and bakes, temperature and time of bakes, and so forth.

(f) Semiconductor data including rating and complete ID for ordering from original supplier (VEDEC number is adequate), and the following:

(1) Rectifiers using silicon diodes.
a. Circuit symbols and piece number.
b. Maximum reverse voltage peak and r.m.s. (complete rectifier circuit).

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c. Maximum load current d.c. amperes (complete circuit).
d. Maximum diode case temperature at design ambient-use for diodes rated against case temperature.
e. Maximum ambient temperature directly adjacent to diodes—use for diodes rated against ambient temperature—no cooling plates.
f. Cooling plate size, thickness and material where so mounted.
g. Air velocity if forced cooled (linear feet per minute).
h. Circuit (for example, single phase bridge, and so forth).
i. Total diodes in circuit.
j. Diodes in series per arm.
k. Diodes in parallel per arm.
l. Description of insulation hardware if insulated from a cooling plate.

(2) Voltage regulator diodes.
a. Circuit symbol(s) and piece number.
b. Range of actual voltage levels which circuit will accept (with or without circuit adjustment from regulator diodes and still give required equipment performance___volts to ____volts.
c. Regulation (including temperature error) permitted in an installed regulator diode under all conditions of operation (assume equipment is initially adjusted for optimum performance) ± ___millivolts.
d. Maximum level of steady-state average reverse current through regulator diode (steady-state shall be considered any time in excess of 30 seconds): ____ma.
e. Maximum transient reverse current under any operating condition___ma for___sec. duration and ___frequency of occurrence.
f. Maximum ambient temperature at diode (no cooling plate).
g. Maximum case temperature of diode (complete for diodes rated against case temperature).
h. Cooling plate size, thickness and material where used.
i. Air velocity if forced cooled (linear feet per minute).
j. Total diodes in circuit.
k. Diodes in series per circuit element (complete where more than one circuit element is used in the voltage regulator such as in a bridge type regulator circuit).
l. Description of insulation hardware if insulated from a cooling plate.

(3) Voltage reference diodes (see regulator diodes also).
a. Circuit symbol(s) and piece number.
b. Range of reference voltage levels which circuit will accept (with or without circuit adjustment) from the voltage reference diode(s): ____volts to ____volts.
c. Error permitted in an installed voltage reference diode under all conditions of operation: ± ____millivolts.
d. Circuit design current through voltage reference.
1. Nominal or average value:____ma.
2. Regulation about nominal value: ± ___ma.
e. Maximum ambient temperature (include internal heat rise above outside design ambient): ___a.
f. Number of voltage reference elements in series: (Operation in parallel is not permitted.)

(4) Transistors.
a. Circuit symbol(s) and piece number.
b. Maximum collector to emitter voltage ___volts.
c. Maximum collector to base voltage ___volts.
d. Maximum collector current ___ma.
e. Maximum base current ___ma.
f. Maximum collector power dissipation ___mw.
g. Maximum ambient temperature (include internal heat rise at transistor at 50°C., design ambient ____°C.
h. Maximum case temperature of transistor at 50°C. equipment design ambient and any condition of required equipment operation ____°C. (Where rating is based on case temperature.)

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i. Cooling plate size, thickness and material where so mounted.
j. Description of insulating hardware if insulated from a cooling plate.

(5) Silicon controlled rectifiers:
a. Circuit symbol(s) and piece number.
b. Nominal working peak forward and reverse voltages.
c. Repetitive peak forward and reverse voltages (maximum).
d. Non-repetitive peak forward and reverse voltages (maximum).
e. False firing, or self-firing, restrictions.
f. Repetitive peak forward current (nominal and maximum values).
g. Surge current (peak for non-repetitive value occuring within the conduction portion of a single cycle but after turn-on has occurred, effective amplitude and cycles duration for multicycle surges, or amplitude and duration for sub-cycle pulse duration).
h. Peak current and maximum rate of current rise during turn-on.
i. Minimum time from the instant of voltage reversal to the instant when voltage is re-applied in the forward direction.
j. Peak forward gate current (minimum and maximum, per device).
k. Peak forward gate voltage (minimum and maximum, per device).
l. Peak reverse gate voltage (maximum, per device).
m. Maximum gate power dissipation (average and peak, per device).
n. Maximum device case temperature at design ambient.
o. Cooling plate size, thickness and material, where so mounted.
p. Air velocity if forced cooled (LFM).
q. Circuit.
r. Total devices in circuit.
s. Devices in series per arm.
t. Devices in parallel per arm.
u. Insulation hardware if insulated from a cooling plate.

(6) Selenium surge suppressors:
a. Identify the suppressor manufacturer's technical data bulletin applicable to each product certified for the application. For each application indicate which product is installed in the equipment and which is the alternate product that may be used for replacement purposes, when desired.
b. Provide the following product rating data:
1. Stack rms voltage (steady-state).
2. Stack clamping voltages (new and aged).
3. Peak discharge current (corresponding to clamping voltage).
4. Polarization (polarized or non-polarized).
5. Maximum energy (watt-seconds) at rated clamping voltage (aged) for single pulse (non-repetitive) current.
6. Maximum pulse duration (microseconds) for repetitive pulses at supply frequency (specify cycles per second) and amplitude equal to clamping voltage (aged).
7. Maximum ambient temperature at which rated values in b, 1 through 5 apply.
8. Maximum operating temperature.

(g) Magnetic amplifier operating characteristic curves:

(1) Plot of amplifier output versus control winding ampere turns.
(2) Schematic diagram of test circuit.

(h) Sketch of terminal boards used for connection of input and output circuits and any external control connections which may be used.

(i) Schematic circuit diagram. In preparation of the schematic, emphasis shall be placed on simplicity and ease of understanding of circuit operation. Physical placement of components and connecting wiring may be ignored in the interest of simplicity and clarity of this diagram. Each part shall be identified by symbols (R-resistor, L-reactor, and other similar symbols) consistent with connection diagram, list of material and description of operation. Terminals used for connections external to the power supply shall be identified and labeled as to purpose. A note giving input voltage, phases and frequency shall be placed adjacent to input

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terminals. A note giving output voltage adjustment range and rated output current or power shall be placed adjacent to output terminals.

(j) Connection diagram showing components and wiring in approximately the relative physical positions which they occupy in the equipment. Leads of components, hookup wire lengths used for interconnection, components, terminals and subassemblies shall be marked. Hook-up wire size need not be shown. Markings shall be consistent with component descriptions given elsewhere. Notes labeling terminals for external connections shall be provided on connection diagram the same as on schematic diagram.

(k) Description of operation - a brief description of operation of the circuit elements to enable approval authorities and operators to understand circuitry.

(l) Briefly presented operating instructions.

3.9.1.1 Data specified in 3.9.1 shall be presented on a maximum of three sheets. Approval for use of a greater number of sheets shall be obtained from the bureau of agency concerned. When manuals are required by the contract or order, data of 3.9.1 (e) through 3.9.1 (g) shall not appear on master drawings, but shall be presented in the manuals. In those cases where master drawings are submitted to the approval authority prior to submittal of manual, the data of 3.9.1 (e) through 3.9.1 (g) may be submitted in any convenient form.

3.9.2 Preliminary drawings.- Preliminary drawings representing preliminary preparation of master drawings may be submitted for design approval in any convenient form, provided data is complete.

3.9.3 Installation drawing.- A separate installation drawing showing data of 3.9.1 (a), 3.9.1 (b)(1), 3.9.1 (b)(2), 3.9.1 (d) and 3.9.1 (h) shall be provided. The installation drawing shall include interconnection information if the power supply system consists of more than one distinct assembly or enclosure.

3.10 Manuals.- Manuals shall be furnished for all class B power supplies. Manuals shall not normally be furnished for class NR power supplies. Manuals shall be furnished for class NR power supplies only when specified (see 6.1). When manuals are furnished, one manual shall be packaged with each equipment. Manuals shall conform to type II of MIL-M-15071 with content and arrangement as follows:

(a) Chapter 1 - General information:

(1) Manufacturer's type and catalog or model number.
(2) Navy designation and specification.
(3) Federal stock number of equipment (if any).
(4) Dimensions.
(5) Weight.
(6) Input power requirements and heat dissipation.
(7) Brief discussion of what the equipment is and what it does.

(b) Chapter 2 - Detailed description and principle of operation:

(1) Detailed description of function of each part. Include theory of operation of parts such as rectifiers, controlled rectifiers, magnetic amplifiers, and other similar parts. Discuss location of each part and provide one or more photographs.
(2) Discussion of how parts described in (b)(1) function together to provide equipment performance.

(c) Chapter 3 - Operating instruction:

(1) Safety precautions.
(2) Start up.
(3) Adjustments:
  a. Output voltage
(4) Use of remote control devices, if any.
(5) Shut-down.

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(d) Chapter 4 - Installation:

(1) Installation instructions (mechanical). Include a sketch. (Reduced size installation drawing may be inserted here.)
(2) Connection instructions (electrical):
a. Input - give input current.
b. Output - give output current. If receptacles for portable cables are used, note polarity.
c. Remote control connections, if any.
(3) Check out before use.

(e) Chapter 5 - Maintenance:

(1) Preventive maintenance
a. Cleaning
b. Inspection
c. Lubrication
d. Placing in service after storage
(2) Corrective maintenance
a. Trouble shooting charts
b. Tables and illustrations of measurements of voltage, current, resistance and waveform at significant test points in the circuitry
(3) Adjustment after parts replacement

(f) Chapter 6 - Parts identification:

(1) Parts tabulation
a. Name, piece number and schematic symbol of part.
b. Equipment manufacturer's part number.
c. Part manufacturer's identification.
d. Federal stock number of part.
(2) Reduced size provisioning list Bureau of Ships mechanical electrical equipment (form 4786A may be used for this part list.).

(g) Chapter 7 - Test data:

(1) Performance (regulation steady state and transient; ripple; efficiency input current) data at no, half and full load, normal input voltage and frequency.

(h) Drawings - reduced size master drawing inserts.

(i) Appendix - inserts on special components.

3.11 Repair parts.-

3.11.1 Repair parts.- Onboard repair parts shall be processed in accordance with MIL-P-15137 and furnished either on a "set" basis or on a consolidated "shipworth" basis as specified (see 6.1).

3.11.1.1 Repair parts per set.- Onboard repair parts furnished with equipments may be specified by "sets". When specified by "sets" each "set" of onboard repair parts shall consist of the following component parts, as modified by 3.11.1.2. and 3.11.1.3.

(a) One complete replacement of all contacts and springs (as applicable to contactors, relays, switches, rheostats, thermostats, push buttons, and similar items) except for overload devices furnished in accordance with MIL-R-2033, MIL-C-2212 or MIL-C-2174 (see 3.11.1.2).

(b) One complete switch of each type and size, if Bureau of Ships design.

(c) Resistor units (see 3.11.1.3).

(d) Capacitor units (see 3.11.1.3).

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(e) One semiconductor device of each type and size and, where matching is required, matched sets of devices shall be furnished.

(f) If potted assemblies or components are used, one potted assembly or component of each type and size shall be furnished.

3.11.1.2 If any of the repair parts listed herein form part of an assembly, or subassembly and is not normally a replaceable part of the assembly or subassembly then the complete assembly will be furnished in lieu thereof as a repair part. Also, in lieu of a complete set of contacts and springs (in such components as switches, thermostats, relays, contactors, rheostats, push buttons, and similar items), the contractor may furnish the complete component. If the complete component is furnished, the quantity shall be not less than 10 percent of the quantity of components of that type and size installed.

3.11.1.3 The quantity of resistor units and capacitor units of each type and size furnished in a set of repair parts shall be not less than 10 percent of the quantity of units of that type and size installed in one equipment. When two or more identical elements are assembled to form a complete unit, the quantity of elements of each type and size furnished in a set of repair parts shall be not less than 10 percent of the quantity of that type and size installed in one equipment, and in no case shall it be less than the number of identical elements connected in a circuit arranged so that failure of one element will result in overloading and progressive failure of the remaining elements.

3.11.2 Repair parts furnished on a consolidated "shipworth" basis. - When onboard repair parts are to be consolidated and furnished on a "ship" basis the number of parts to be supplied shall be based on the total number of identical components (contactors, relays, rectifying elements, switches, resistor units, and similar items) installed in which the part is used. The parts and complete components to be furnished as repair parts shall be as shown in table I, as modified by 3.11.1.2 and 3.11.2.1.

Table I - Parts and complete components.

Number of components installed	Parts to be furnished				
	Complete replacement of contacts and springs	Switches (Bureau of Ships design)	Resistor and capacitor units	Semiconductor devices	Potted assemblies
1 to 3	1	1	1	1	1
4 to 10	2	1	2	2	1
11 to 20	3	2	3	3	2
21 to 40	4	3	4	4	3
41 to 65	5	3	5	5	3
66 to 100	6	4	6	6	4
101 and over	8	4	8	8	4

3.11.2.1 When two or more identical resistor or capacitor elements are assembled to form a complete resistor or capacitor unit, the quantity of elements of each type and size furnished as repair parts shall be not less than the number of identical elements connected in a circuit arranged so that failure of one element will result in overloading or progress in failure of the remaining elements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Qualification tests 1/.- Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the examination and tests specified in table II and in the individual detail specification.

4.2.1 Power supplies to be submitted for qualification approval.- Qualification approval may be obtained as specified in 4.2.1.1 through 4.2.1.3.

4.2.1.1 Individual types.- One power supply for type(s) covered by an individual detail specification for which qualification approval is desired shall be submitted and subjected to the qualification inspection specified in table II. Qualification approval will be granted only for the power supply or power supplies specified in the individual detail specification.

4.2.1.2 Class R and NR.- Where qualification approval is desired for all class R and class NR power supplies, one power supply of the type specified in MIL-P-15736/4 shall be submitted and subjected to the qualification inspection specified in table II. Approval of power supplies conforming to MIL-P-15736/4 will include approval of all class R and NR power supplies.

4.2.1.3 Class NR.- Where qualification approval is desired for all class NR power supplies, one power supply of the type specified in MIL-P-15736/2 shall be submitted and subjected to the qualification inspection specified in table II. Approval of power supplies conforming to MIL-P-15736/2 will include approval of all class NR power supplies.

4.2.2 Prior to test authorization, the manufacturer shall submit detailed drawings containing all information as required for master drawings and any other specific information, as requested by the Bureau of Ships, for determining complete compliance with requirements of applicable specifications. Manufacture of equipment for tests should not begin until receipt of manufacturing drawing approval by the Bureau of Ships (except for prototype breadboard samples which may be built anytime by the manufacturer to prove certain operating features and circuitry). Also, prior to test authorization, the manufacturer shall submit evidence that the equipment conforms to performance requirements of applicable specifications and shall officially request in writing that qualification tests be conducted.

Table II - Qualification routine, selective and periodic inspection

Description	Routine	Selective	Periodic	Qualification	Reference paragraph	
					Requirement	Inspection
General examination	X	X	X	X	- - -	4.4
Efficiency	X	X	X	X	3.5.2.9	4.5.1
Insulation resistance	X	X	X	X	3.7.1.1	4.5.2
Effectiveness of enclosure	- - -	- - -	X	X	3.6.1	4.5.3
Heating	- - -	- - -	X	X	- - -	4.5.4
Weight	- - -	- - -	X	X	3.6.4	4.5.5
Inclined operation	- - -	X	X	X	3.5.3.1	- - -
Shock	- - -	- - -	X	X	3.5.3.1	4.5.6
Vibration	- - -	- - -	X	X	3.5.3.1	4.5.7
Starting	- - -	X	X	X	3.6.5.2.1	4.5.8
Phase balance	- - -	- - -	X	X	3.5.2.3	4.5.9

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3 and 6.3.1).

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Table II - Qualification routine, selective and periodic inspection (Cont'd)

Description	Routine	Selective	Periodic	Qualification	Reference paragraph Requirement	Reference paragraph Inspection
D.c. output and special performance requirements	- - -	X	X	X	3.5.2.4	4.5.10
Creepage and clearance distance	- - -	- - -	X	X	MIL-E-917	4.5.11
Dielectric	X	X	X	X	3.7.1.2	4.5.12
Airborne noise (see note 1)	- - -	- - -	X	X	3.5.4.1	4.5.14
Structureborne noise (see note 2)	- - -	- - -	- - -	- - -	3.5.4.2	4.5.14
Environmental	- - -	- - -	- - -	X	3.5.3.3	4.5.13

NOTES:

1. Airborne noise is a periodic and qualification inspection; however, specific contracts or orders may require units selected for airborne noise inspection to be so selected in accordance with MIL-STD-740.
2. When specified (see 6.1), structureborne noise inspection will be required on units selected in accordance with MIL-STD-740.

4.3 Routine, selective and periodic inspection. -

4.3.1 Routine inspection. - Each power supply offered for delivery shall be subjected to the routine inspection specified in table II. If any power supply fails in any routine examination or test, the manufacturer shall take corrective action to enable the equipment to conform with the requirements of this specification. Reworked power supplies shall be re-examined and retested to demonstrate that the equipment conforms to the requirements of this specification. Reworked power supplies which again fail routine examination and tests shall be rejected.

4.3.2 Selective inspection. -

4.3.2.1 Definition of a lot. - All equipment of the same type presented for quality conformance acceptance inspection at one time shall be considered a lot. The lot may include the entire quantity required by one or contracts or it may be the production of any convenient time period as agreed upon between the contractor and the Government.

4.3.2.2 Sampling for selective inspection. - A random sample of equipments shall be selected from each lot in accordance with table III, and shall be subjected to the selective inspection of table II. Failure of any unit in the selected sample to pass any of the selective examination or tests shall be cause for rejection of that lot.

Table III - Sampling for selective inspection

Lot size	Sample size
1 to 5	1
6 to 8	2
9 to 10	3
16 to 40	5
41 to 65	7
66 to 110	10
111 to 180	12
181 to 300	15
301 or more	Not less than 5 percent

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If a lot is rejected due to failing the selective inspection no further lots will be accepted for inspection until the manufacturer has taken corrective measures and has demonstrated, to the satisfaction of the Government representative that these measures will enable the equipment to conform to the requirements of this specification. Rejected lots may be reworked and resubmitted but shall be kept separate from new lots. Failure of any units of a resubmitted lot in any examination or test which caused initial rejection shall be cause for final rejection of the lot.

4.3.3 Periodic inspection.- The first power supply of each design offered for delivery shall be subjected to the periodic inspection of table II. A periodic inspection shall be required after any change in design affecting performance characteristics. Only one periodic inspection of a specific design of equipment will be required on a particular contract or order. In case of subsequent contracts or orders for identical equipment, the periodic inspection shall be repeated if 18 months or more have lapsed between the date of the last previous periodic inspection and the date of the subsequent contract or order. However, shock and vibration tests are required only once on any specific design of equipment. If the power supply subjected to the periodic examination and tests fails in any examination or in any test, no power supplies shall be accepted for routine or selective inspection until the manufacturer has taken corrective action on that power supply and on every other power supply to be offered for delivery to assure that the equipment complies with the requirements of this specification. The examination or test failed and any other periodic examination and test required by the Government shall be repeated to demonstrate that the corrective action will enable the equipment to conform to the requirements of this specification. Only one re-examination and retest will be permitted. The re-examination and retest shall be made within such time after the first examination and test as stipulated by the Government. Failure to correct defects and successfully pass the re-examination and retests shall be cause for final rejection of the equipment.

4.4 General examination.- Items on major parts and completed units shall be examined as specified in 4.4.1 and 4.4.2 to determine conformance with the requirements of this specification.

4.4.1 Major parts.- Examination shall be as follows:

(a) To determine conformance with the applicable specification and that the size, type and rating are as shown on the approved drawing for the following items:

(1) Resistors and rheostats.
(2) Instruments, instrument transformers, shunts and other associated accessories.
(3) Motor starters and associated pushbuttons.
(4) Circuit breakers.
(5) Capacitors.
(6) Switches.
(7) Semiconductor devices.
(8) Terminal boards.
(9) All laminated materials, such as glass mica glass (GMG), glass silicon glass (GSG); used for terminal boards; sub-panels; and so forth.

(b) All materials used in the construction of reactors, magnetic amplifiers, and transformers are as shown on the approved drawing and comply with the material requirements as shown on the approved drawing.

(c) All windings of reactors, magnetic amplifiers, and transformers were given varnish impregnation in accordance with the procedures shown on the approved drawing.

(d) No black varnish, glyptol, paint, or other treatment was given to the insulation of reactors, magnetic amplifiers, and transformers.

(e) The resistance of all windings of all reactors, magnetic amplifiers, and transformers is as shown on the approved drawing. Ratio of turns is as specified on approved drawing.

(f) The equipment enclosures are as shown on the approved drawing.

4.4.2 Completed unit.- Examination shall be conducted to determine that:

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(a) All components are mounted as shown on the approved drawing.

(b) All parts which require servicing, repair, replacement, or periodic adjustment during the life of the equipment are readily accessible.

(c) All components are marked with identifying symbols and these symbols agree with those used on the approved drawing.

(d) All terminals of all components and terminal boards are marked with identifying letters and numbers, or both, and this identification agrees with those used on the approved drawing.

(e) Wire used for interconnecting components is of the type shown on the approved drawing.

(f) All wiring is neatly formed into groups and laced with non-flammable glass cord and supported or clamped in a manner which will prevent chafing of the insulation due to vibration and shock.

(g) Cable clamps used are of non-flammable material.

(h) There are no splices in any of the wires.

(i) Wire groups from hinged doors and panels are formed and clamped in a manner that sharp bends do not occur with the panel or door in either the open or closed position.

(j) Sufficient slack in wiring is allowed so that the weight of the harness is not supported by the terminal connections and so that at least two replacements can be made if lugs are clipped off at the wire end.

(k) Cable entrance provisions are as shown on the approved drawing.

(l) Sufficient space is provided to bring external cables through these cable entrances, support them within the enclosure, and make connections at the appropriate terminals.

(m) Wherever wires run through holes in metal partitions or chassis, grommets are provided for mechanical protection.

(n) Wires are not bent around sharp corners which may injure the insulation.

(o) All wires are connected by either bolted or soldered connections.

(p) All bolted connections are provided with locking devices.

(r) Both ends of all wires are marked with designations as shown on the approved drawing.

(s) Hinged doors and panels do not bind when opening.

(t) A door stop and positioning device is furnished and functions satisfactorily.

(u) All identification and instruction plates are furnished as shown on the approved drawing.

4.5 Tests.-

4.5.1 Efficiency.- The equipment shall be loaded with resistive load during the test. Meters shall be inserted in the a.c. input for reading voltage, current and wattage and also in the d.c. output for measuring voltage and current. Readings shall be recorded at no load, half-load and full-load at rated input voltage and frequency. Efficiencies shall be calculated at half and full load by means of the following formula:

$$\text{Percent efficiency (overall)} = \frac{\text{D.c. volts x d.c. amperes x 100}}{\text{A.c. watts input}}$$

These efficiencies shall equal or exceed the manufacturer's rated efficiencies.

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4.5.2 Insulation resistance.- Insulation resistance shall be as specified in 3.7.1.1. The measurement of insulation resistance shall be made with all circuits of equal voltage above ground connected together. Circuits or groups of circuits of different voltage above ground shall be tested separately. Insulation resistance shall be measured with an insulation-resistance-indicating meter conforming to type GC of MIL-O-16485. The time of test voltage application shall be not less than 60 seconds. Measurements shall be made at any convenient ambient temperature with all circuits cold. The relative humidity and ambient temperature shall be recorded. Test conditions of 3.7.1.1.1 shall apply.

4.5.3 Effectiveness of enclosure.- The equipment enclosure shall be tested in accordance with MIL-E-2036 for the corresponding type of enclosure except for types dripproof and dripproof protected. Dripproof and dripproof protected enclosures shall be examined to ascertain that the enclosure effectively performs its specified function.

4.5.4 Heating.- Heating tests shall be made under conditions equivalent to normal operating conditions at maximum rated voltage, maximum rated load, and the duty specified. The test methods to be employed and the precautions to be observed shall be as specified in 4.5.4.1 through 4.5.4.4.

4.5.4.1 Assembly and mounting of equipment.- Heating tests shall be made only with the equipment completely assembled and mounted in the manner for which it is designed. Barriers shall be placed adjacent to the enclosure, at the manufacturer's recommended minimum clearance for adequate ventilation, to simulate shipboard space restrictions. Heating tests shall be made on all coil windings (including power transformers and reactors) and rectifier cells except for those rectifier cells of control circuits which clearly do not contribute to the overall temperature rise within the equipment enclosure and except for those which are tested and given separate qualification approval under its individual specification. For those components which are given exception based on separate qualification approval, it shall be determined that its adjacent ambient within the enclosure does not exceed the rated ambient of the component and that the application rating does not exceed the manufacturer's approved rating at the applied temperature ambient.

4.5.4.2 Method of temperature measurements.- The method of temperature measurement shall be in accordance with MIL-E-917.

4.5.4.3 Duration of test.- The heating test for continuous duty equipment shall be continued until the temperature rise of components that can be observed during test have attained a steady final value. The heating test for intermittent duty equipment shall be continued for the duty cycle specified for the equipment.

4.5.4.4 Method of loading.- The equipment shall be loaded with rated applied input voltage and frequency at full rated output load. The type of load shall be as specified herein or resistive if more convenient.

4.5.5 Weight.- The weight of the equipment and the weight of the onboard repair parts shall be taken and recorded separately. The weight of the equipment shall not exceed the value shown on the applicable approved drawings or the maximum value specified.

4.5.6 Shock.- The high impact shock tests (electrical operation and mechanical damage) shall be conducted in accordance with type A of MIL-S-901.

4.5.6.1 Test features.- Test features shall be as follows:

(a) General.- Type A (for completely assembled apparatus) test shall be performed under light or medium weight classification, as required. In all cases the equipment shall be checked for specified electrical operation following the shock test. The equipment shall be energized during shock test. Shock test may be performed at no load. All class R power supplies and any class NR power supply having automatic operating features or range of manual voltage adjustment shall be tested under load to determine specified operation following shock. Electrical tests as required shall be made to determine satisfactory performance following or during shock test.

(b) Definition of failure to perform principal functions:
(1) Breakage of any parts, including mounting bolts. Minor chipping of parts such as plastic knobs and cases and minor distortion of parts will be permitted where such chipping or distortion cannot in any manner impair operation of the equipment as specified.

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(2) Appreciable distortion of any parts, including enclosure and framework.
(3) A value of insulation resistance lower than that permitted by this specification (see 3.7.1.1).
(4) Low dielectric strength. After shock tests the dielectric test shall be conducted at a voltage equal to 65 percent of the voltage specified in 3.7.1.2. Failure to pass this test shall be cause for rejection.
(5) Failure to pass visual examination. The equipment shall be carefully examined after removing all removable panels and doors to ascertain any mechanical damage. When required, partial disassembly shall be performed to aid in determining any possible damage.
(6) Failure to perform any of the required electrical tests, during and following shock tests. No adjustment or replacement of damaged parts shall be permitted during shock test unless specifically permitted by the bureau or agency concerned.

(c) Mounting.- The equipment shall be mounted on the shock machine in a manner simulating shipboard installations. The specific method or methods of mounting shall be determined as required by MIL-S-901.
(1) Lightweight equipment (up to 400 pounds) shall use the following fixtures:
a. "Standard mounting for bulkhead mounted equipment (type A test LWS)" (fixture 4-A).
or
b. "Standard mounting for deck or platform mounted equipment (type A test LWSM)" (Fixture 4-C).
(2) Medium weight equipment heavier than 400 pounds shall use the following fixtures:
a. "Standard mounting platform for testing equipment on medium weight shock testing machine".
and
b. "30 degree mounting fixture for testing base mounted equipment on medium weight shock testing machine"
or
c. "30 degree mounting fixture for testing bulkhead mounted equipment on medium weight shock testing machine".
(3) Specified mounting - equipment for which both deck and bulkhead mounting is specified shall be tested bulkhead mounted.
(4) Meters - electrical meters may be removed from the equipment during shock tests, but shall be replaced with equivalent dummy weights.

(d) Place at which shock test will be conducted.- At a Government laboratory, at the manufacturer's plant or other testing facility as approved by the bureau or agency concerned.

(e) Number of units to be shock tested.- One equipment shall be shock tested during qualification or periodic tests.

(f) Disposal of shock tested equipment.-
(1) Equipment which has been subjected to HI shock test under qualification tests shall be returned to the manufacturer for his disposal or retained, upon request of the manufacturer, for disposal by the Government.
(2) Equipment which has been subjected to HI shock test under periodic testing shall be accepted as a production unit of the contract or order only under the following conditions:
a. That any damaged parts are replaced.
b. That post shock electrical tests show compliance with specified performance. (The extent of electrical testing, after shock, for specified operation shall be determined as necessary, in each instance and shall be dependent on the nature of required performance and inspection results of shock test).
c. That an identification plate is added identifying the unit as "HI shock tested".
d. That the unit is subjected to the same guarantee by the manufacturer as other production units.
e. That not more than one shock tested equipment of a specific design per contract or order is being offered for acceptance following HI shock tests.
f. Extension of shock test approval. - Acceptance of a particular item of one equipment as HI shock proof may be made on the basis of previous acceptance of similar material as specified in MIL-S-901.

4.5.7 Vibration. - Vibration tests shall be conducted in accordance with type I requirements of MIL-STD-167.

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4.5.7.1 Test feature.- The test features as specified for shock tests (see 4.5.6.1) shall apply also to vibration tests except that standard fixtures may be used throughout, and 30 degree fixtures are not applicable. The unit of equipment selected for vibration test shall be the same unit as selected for HI shock tests.

4.5.8 Starting.-

4.5.8.1 Purpose.- This test shall be made only for equipments employing forced ventilation to ascertain that components will not be damaged by remaining energized in the event of failure of fan or blower or associated motor.

4.5.8.2 Procedure.- The fan motor leads shall be disconnected so that the fan does not rotate. With input power applied to the terminals of the equipment and the control switch actuated to energize the equipment under full load, observation shall be made to ascertain that the equipment remains deenergized or that the equipment becomes automatically deenergized before the temperatures of the semiconductor components exceed normal allowable maximum values.

4.5.9 Phase balance.- Equipment with 3-phase, 3-wire ungrounded input shall be tested at full load for current balance in the a.c. lines. The a.c. line current of each phase shall be measured for r.m.s. measurements.

4.5.10 Performance tests.-

4.5.10.1 Testing conditions.- Normally, performance tests listed herein may be considered to be both necessary and sufficient to demonstrate compliance with requirements of individual detail specifications. Individual detail specifications may contain and require tests and test procedures which modify or supplement the testing requirements herein.

4.5.10.2 Voltage regulation.- Equipment shall be tested under all conditions of voltage, frequency, ambient, and load within the specified rating range. The output voltage shall remain within the specified regulation limits without adjustment other than the initial setting. Voltage regulation tests shall be conducted under the conditions specified in table IV. The first set of readings shall be taken within 2 minutes after the equipment is energized.

(a) The load shall be varied from no load to full load in approximately 5 equally spaced steps under each condition of line voltage, frequency, and ambient temperature listed in table IV.

Table IV - Conditions of line voltage, frequency and ambient temperature.

Input voltage volts	Input frequency c.p.s.	Ambient temperature °C.
Rated (that is, 115, 230, 440 etc.)	Rated (that is, 60, 400 etc.)	50
Rated +5 percent	Rated +5 percent	50
Rated +5 percent	Rated -5 percent	50
Rated -5 percent	Rated +5 percent	50
Rated -5 percent	Rated -5 percent	50
Rated +5 percent	Rated +5 percent	Approx. 23 (room)
Rated +5 percent	Rated -5 percent	Approx. 23 (room)
Rated -5 percent	Rated +5 percent	Approx. 23 (room)
Rated -5 percent	Rated -5 percent	Approx. 23 (room)
Rated	Rated	Approx. 23 (room)

(b) Procedure of regulation runs in (a) (table IV) shall be repeated for voltage settings over the range of specified adjustment. The selected voltage settings shall be at the minimum and maximum limit of specified voltage adjustment and at any other convenient voltage setting approximately half way between these extremes.

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(c) At each condition of input voltage, frequency, and ambient temperature, the output voltage shall be varied slowly over the full range of adjustment at no load and full load to determine that specified range of adjustment is provided.

(d) The maximum and minimum obtainable output voltages shall be taken and recorded for no load and full load conditions at rated input voltage and frequency and at room temperature.

(e) For equipments with remote voltage sensing or automatic d.c. line voltage drop compensation, all of the regulation tests specified in (a) through (c) shall be made under both the condition of maximum specified external line drop and zero line drop (load connections direct to the power supply). Voltage readings shall be taken and recorded simultaneously at the load and at the power supply terminals.

(f) Readings shall be taken and recorded on input voltage and frequency, ambient temperature, output voltage and output current for all of the conditions specified in (a) through (e). Curves shall be plotted of output voltage versus output current for each of the specified voltage regulation test run (see table IV and (b) and (c)).

4.5.10.3 Resolution.- For equipment in which output voltage is adjusted by operation of a selector switch or by connection of transformer taps in a specific manner to yield a specific output voltage value, tests to determine that a specific desired value of voltage (within specified tolerance) can be obtained shall be run. Procedure shall consist of operating selector switch(es) or of connecting transformer leads or taps to all normal positions, recording no load and full load voltage at high, low and rated input voltage and frequency. Data shall be compared with specified performance to determine that specified resolution is obtained.

4.5.10.4 Line voltage transient test.- Tests shall be made to determine conformance with the requirement of 3.5.2.1.1 herein as follows:

(a) Test conditions.- Power supply shall operate at rated output current. Test shall be run at rated input frequency and room ambient temperature.

(b) Test features.- Circuit for producing transients in a.c. voltage shall consist of a variable a.c. supply connected through some known impedance to the power supply. A contactor arranged to insert or remove (short) the impedance shall be provided. In all transient tests the impedances and the contactors shall be single or 3-phase as required. Recording devices shall be as specified in 4.5.10.7.

(c) Procedure.- A.c. supply shall be adjusted so that power supply receives rated voltage. Contactor shall be energized so that series impedance is inserted or removed, as applicable to produce voltage dip or rise. The r.m.s. values of input voltage and current and the output voltage shall be recorded. Tests shall be in accordance with Table V.

Table V - Input transient tests

Power supply voltage (output)	Transient magnitudes, input voltage	
	Dip	Rise
	Percent	Percent
Maximum	10 18	10 18
Minimum	10 18	10 18

4.5.10.5 Operating modes other than voltage regulation.- Where operating and performance features other than voltage regulation are specified, the power supply shall be tested to demonstrate compliance with such specified performance. Typically, such features may include current regulation or adjustment, current limiting, voltage cutoff or limiting and transient performance. Since different equipments, will by individual detail specification, have different expressions of special output and performance requirements,

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the precise method of testing (that is, method of applying load, duration of load, sequence of operation and similar methods) may necessarily vary and in each case shall be tailored to best satisfy the condition of specified operation. Individual detail specifications shall set forth test procedures for special performance requirements.

4.5.10.6 Ripple voltage.-

4.5.10.6.1 Method of measuring ripple.- Method of measuring ripple voltage shall be as follows:

(a) Measure effective (r.m.s.) value of the ripple voltage:

(1) By sensitive electrodynamometer type a.c. meter or thermal type meter in series with a blocking capacitor.
(2) By vacuum tube voltmeter, calibrated in r.m.s. values, in series with a blocking capacitor.
(3) By oscilloscope. The peak-to-peak value of the ripple voltage shall be recorded and considered to be equal to three times the r.m.s. ripple voltage.

(b) Measure average value of the pulsating d.c. output voltage with a permanent magnet moving coil meter or rectifier type meter.

(c) Calculate percent ripple:

$$\text{Percent ripple voltage} = \frac{\text{r.m.s of ripple voltage}}{\text{average of pulsating voltage}} \quad X100$$

4.5.10.6.2 Conditions for measuring ripple.- Measurements specified in 4.5.10.6.1 shall be made at no-load, one-half load and full load, at rated input voltage and, for 3-phase input power supplies, with input phase voltages unbalanced as follows:

Test condition	ØA	1ØB	ØC
1	Rated	Rated	Rated -3 percent
2	Rated	Rated -3 percent	Rated -3 percent
3	Rated -3 percent	Rated -3 percent	Rated
4	Rated -3 percent	Rated	Rated
5	Rated -3 percent	Rated	Rated -3 percent

4.5.10.7 Electrical test instruments.- Indicating meters used for measurement of voltages and currents (input and output) during tests shall have maximum error of indication of one-quarter percent of full scale indication. Meters used for measurement of ripple content of output voltage shall have maximum error of indication the same as above, except where vacuum tube voltmeters are used. Where VTVM's are used for ripple measurements, the maximum error of indication shall not exceed 5 percent. Transient performance shall be recorded with oscillographs equal or similar to Heiland "Visicorder" with galvanometers of appropriate frequency response, or with oscilloscope and camera combination equal or equivalent to Tektronix, Inc. Type 536.

4.5.11 Creepage and clearance distances.- Creepage and clearance distances shall be demonstrated by actual measurement to be in accordance with MIL-E-917.

4.5.12 Dielectric tests.-

4.5.12.1 General.- Dielectric tests shall be performed after the examination (see 4.4) and all of the tests specified in 4.5.1 through 4.5.11 have been completed. If the insulation resistance of the windings is known to be lower than specified, due to dirt, moisture or damage to windings this shall be remedied before the application of the dielectric test voltage.

4.5.12.2 Test voltage.- The frequency of the testing voltage shall be not less than 60 cps and shall approximate a true sine wave. The value of test voltage shall be as specified in 3.7.1.2 and shall be applied continuously for a period of 1 minute.

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4.5.12.3 Measurement of test voltage.- The measurement of the voltage used in dielectric tests shall be made by the voltmeter method whereby the meter derives its voltage from the high-voltage circuit either directly or by means of a voltmeter coil placed in the testing transformer, or through an auxiliary ratio transformer.

4.5.12.4 Points of application.- The test voltage shall be successively applied between each electric circuit and all other electrical circuits and grounded metal parts not electrically connected to it. Semiconductor components may be shorted out or disconnected for the dielectric test.

4.5.13 Environmental test.-

4.5.13.1 General.- Environmental test shall be conducted to determine:

(a) If all transformers and reactors have received proper insulation treatment and if the manufacturer's treatment process is completely effective to give the desired design life.

(b) If all components and hardware have received the proper corrosion-resistant treatment and if the assembly procedures of the manufacturer in any way destroy the complete effectiveness of the treatment.

(c) If any miscellaneous weaknesses exist in construction.

(d) If performance is affected by exposure to high humidity.

4.5.13.2 Procedure.- All transformer and reactor windings shall be disconnected as necessary from other components to permit accurate measurements of insulation resistance and dissipation factor of the winding and ground insulation. Initial measurements of insulation resistance and dissipation factor shall be taken as well as the temperature and relative humidity of the surrounding air. The equipment shall next be conditioned for a period to arrive at a dry condition. Conditioning shall be obtained by placing the equipment in a heat chamber or drying oven for a period of 6 hours with temperature maintained at 105°C. At the end of this period the equipment shall be removed and after cooling (within 8 hours) the insulation resistance and dissipation factor shall again be measured. The equipment shall then be immediately placed in the humidity chamber for a period of 1 week (approximately 168 hours). While under humidification, daily measurements of ambient temperature, insulation resistance and dissipation factor shall be made. At the end of the seventh day humidification period and within 5 minutes after removal from the humidity chamber the insulation resistance and dissipation factor shall be measured. Following measurements the equipment shall be carefully examined for any signs of material damage and corrosion. The extent of any observed damage or corrosion shall be noted and cause determined if feasible. After examination, any disconnections or disassembly shall be reverted to original condition and the equipment shall be operated electrically to determine if the electrical performance has in any way been affected. Also following operation, insulation resistance shall be measured and recorded between isolated circuits and between circuits and ground.

4.5.13.2.1 Recommended test apparatus.- Recommended test apparatus shall be as follows.

(a) The humidity chamber can be made of steel or by placing a transparent plastic film having a low moisture permeability over a steel framework. The top of the chamber shall be slanting or peaked (or otherwise arranged) so that the excess condensate does not drip on the equipment. The humidity shall be maintained at 100 percent RH by electrically heating an open pan of water within the chamber. The amount of exposed surface of the water shall be sufficient to produce minute droplets of condensate on the equipment and insulation surfaces under test. However, the amount of condensate shall be controlled so as not to produce puddles or streams of water on the insulation surfaces. The equipment door or panels shall be opened or removed to permit free access of the humidified air to all parts and surfaces.

(b) The insulation resistance measuring equipment shall be similar and equivalent in accuracy to the General Radio Megohm Bridge, type 544-B, 500 volt d.c. with a range from 0.1 to 1,000,000 megohms.

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(c) The dissipation factor measuring equipment shall be similar and equivalent in accuracy to the General Radio Capacitance Test Bridge, type 1611-A, 60 cycles with a range of dissipation factor from 0 to 50 percent.

4.5.13.2.2 Methods of measurement. - Methods of measurement shall be as follows:

(a) The insulation resistance measurement shall be made using a potential of 500 volts, d.c. applied for a 1 minute period. The insulation resistance shall be measured on all transformers and reactors. Measurements shall be made and recorded between each winding and ground and between insulated windings.

(b) The dissipation factor shall be measured with the capacitance test bridge with values read directly in percent from the bridge. Dissipation factor shall be measured on all transformers and reactors. Measurements shall be made and recorded between each winding and ground and between insulated windings.

NOTE: Due to the need for exposing copper for the connections to the windings, a leakage path is produced by humidification over the various insulation surfaces between the base copper and the other parts of the equipment that are otherwise insulated from the copper circuit. Therefore, leads should be held separated as much as possible and covered with a nonwetting grease which is wiped off with a clean dry rag upon removal from humidification. All measurements shall be made either with equipment in the humidity chamber or within 5 minutes after removal of the equipment from the humidity chamber.

4.5.14 Airborne and structureborne noise. -

4.5.14.1 Measurements. - Airborne and structureborne noise measurements shall be made and recorded in accordance with MIL-STD-740 with the power supply operating at full load at rated input voltage and frequency.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Power supplies. -

5.1.1.1 Preservation and packaging. - Preservation and packaging which may be the suppliers commercial practice shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the uniform freight classification ratings, rules and regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the suppliers commercial practice. The information shall include nomenclature, Federal stock number or manufacturers part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. - Onboard repair parts shall be preserved and packaged level A; packed level C and marked levels A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. - The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

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5.2.1.1 Power supplies. -

5.2.1.1.1 Preservation and packaging. -

5.2.1.1.1.1 Level A. - The equipment shall be preserved and packaged level A in accordance with MIL-P-16298.

5.2.1.1.2 Packing. -

5.2.1.1.2.1 Levels A and B. - The basic equipment shall be packed level A or B as specified in the contract or order, in accordance with MIL-P-16298.

5.2.1.1.3 Marking. - In addition to any special marking required by the contract or order, marking for shipment shall be in accordance with MIL-P-16298.

5.2.1.2 Repair parts. - Repair parts shall be preserved, packaged, packed and marked for the level of shipment as specified, in the contract or order, in accordance with MIL-P-16298.

5.2.1.3 Manuals. - Manuals shall be packaged, packed and marked as specified, in the contract or order, in accordance with MIL-M-15071.

5.2.1.4 Drawings. - Drawings shall be packaged and packed in accordance with MIL-D-963.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification and the individual detail specification.

(b) Class required (see 1.2) (see note)

NOTE: Equipment classification shall normally be based on the requirements of the application. Use of class NR power supplies is desirable where practical, because of their generally lower cost and greater simplicity. The factors which determine whether a class NR power supply can be used in a specific application are given in 6.2.

(c) Detail requirements (see 3.1)

(d) Electrical rating and performance (see 3.5.2):
- (1) Ac input voltage (see 3.5.2.1)
- (2) Input frequency (see 3.5.2.2)
- (3) Phases and balances (see 3.5.2.3)
- (4) Dc output (see 3.5.2.4).
  - a. Output voltage adjustment range and resolution
  - b. Output voltage regulation required
    1. With input voltage changes
       - a. Steady state
       - b. Transient
    2. With output load current changes.
       - a. Steady state
       - b. Transient
         1. Maximum permitted initial excursion of output voltage on load application and rejection.
         2. Time to recover steady value.
         3. Maximum permitted overshoot during recovery.
       - c. Remote regulation (d.c. line drop compensation
         1. Whether remote sensing is required (see 3.5.2.4.1).
         2. Maximum voltage drop (at rated output current) between power supply and remote point at which voltage is to be regulated.

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(e) Permissible ripple content) (see 3.5.2.5).

(f) Whether remote control of output voltage is required.

(g) Rating (normally given in output current (see 3.5.2.5.).
(1) Continuous
(2) Intermittent, if applicable.
a. Curr nt
b. Time
(3) Degree and type of protection (for example, fuses) (see 3.5.2.8).

(h) Current limiting characteristics (if any) required.

(i) Duty, if other than continuous (see 3.5.2.5)

(j) Switching required (if any)
(1) Input
(2) Output
(3) Remote control, if required.

(k) Meter and meter accessories required (see 3.8.3):
(1) Type and size
(2) Range
(3) Special marking, if applicable.

(l) Structureborne noise limits required (see 3.5.4.2. and note 2, table II).

(m) Radio frequency (see 3.5.5)

(n) Ventilation (see 3.6.5)

(o) Mounting (see 3.6.2)

(p) Ambient temperature, if different from 3.5.3.2.

(q) Enclosure type, if other than dripproof protected (45 degrees) (see 3.6.1).

(r) Maximum size and weight (see 3.6.4):
(1) For submarine equipment, whether equipment must pass submarine hatch, watertight door, or both (see 3.6.4.1).

(s) Type of load (see 3.5.2.6) Note. For battery charging, specify type of battery, if other than lead acid.

(t) Whether manuals are required (see 3.10).

(u) Repair parts (see 3.11.1).

(v) Preservation, packaging, packing and marking required, if other than specified in 5.1 (see 5.2).

(w) Exceptions, if any.

6.2 Class NR power supplies. - The following factors are to be considered in determining if a class NR power supply is suitable for a given application:

(a) A basic assumption is that a typical class NR power supply consists of a transformer with primary or secondary (or both) taps which may be connected in various ways to yield various values of output voltage, a rectifier assembly or stack, and a bleeder or dummy load resistor. Also, included may be fuses, switches, output filter components, and so forth.

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(b) Since no automatic regulating devices are used, output voltage will change with change in input voltage and also with change in output current.

(c) Output voltage will decrease as output current increases, because of voltage drop in transformer, internal wiring and rectifier assembly. A representative figure for this drop is 6 percent of no-load voltage for a change from zero to full rated load. By proper design and choice of components, this dro can be made nearly linear.

(d) Input voltage may vary from 105 percent to 95 percent of rated value. Output voltage of class NR power supply varies with the input voltage.

(e) Therefore, for example, a class NR power supply, adjusted to a desired value at no load, with 105 percent rated input voltage applied would have an output voltage 16 percent lower than desired when fully loaded with 95 percent rated input voltage applied. In any case of application of a class NR power supply an optimum adjustment can be made, using as variables both the output voltage adjustment scheme (normally transformer taps) provided and the operating conditions for which the power supply is adjusted to the desire value.

(f) If a load circuit energized by a class NR power supply draws a constant load current, then only input voltage change need be considered in estimating change of voltage during operation.

6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List 15736, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington, D.C., 20360 and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, 19120 Pennsylvania.

6.4 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

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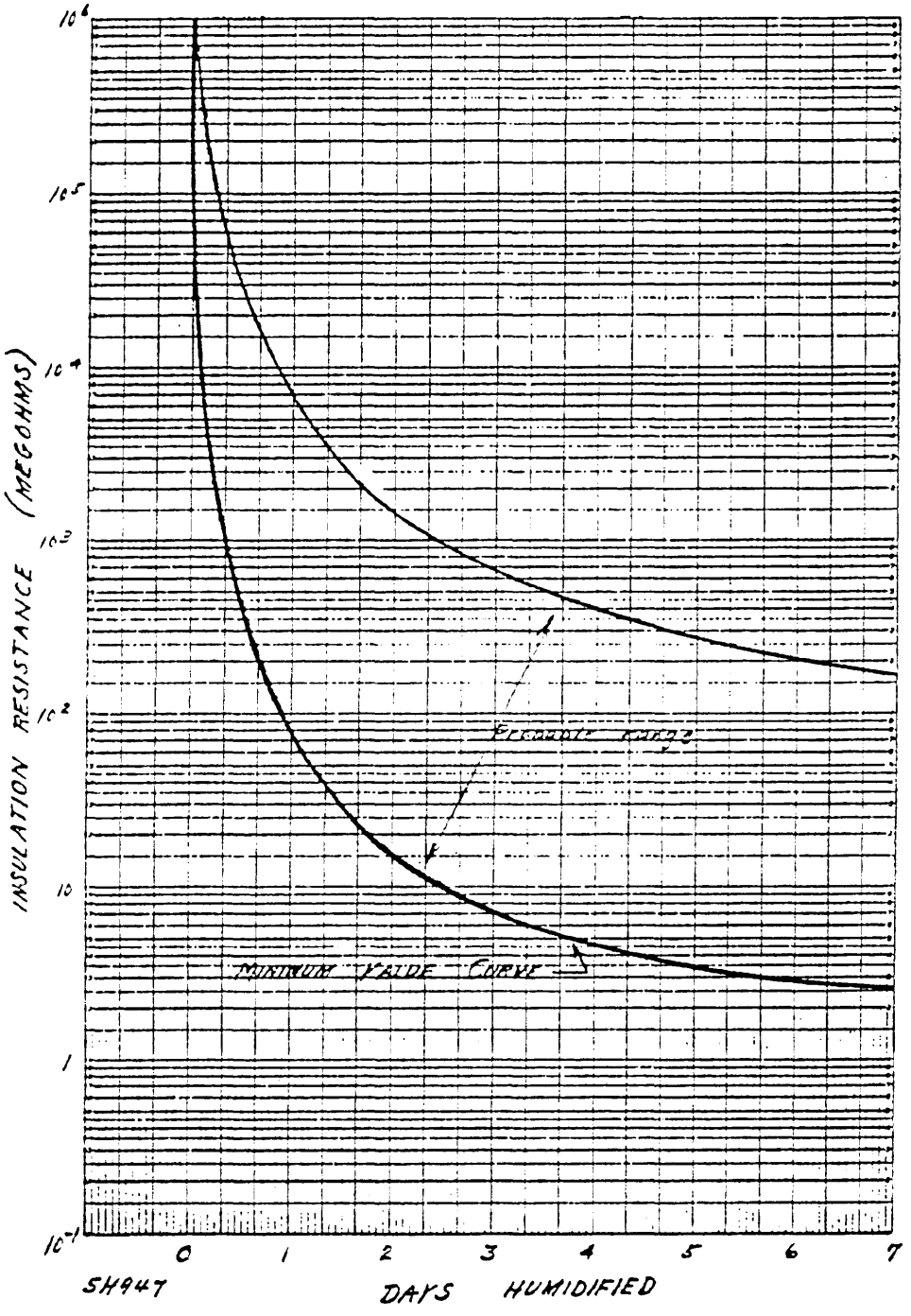


Figure 1 - Insulation resistance.

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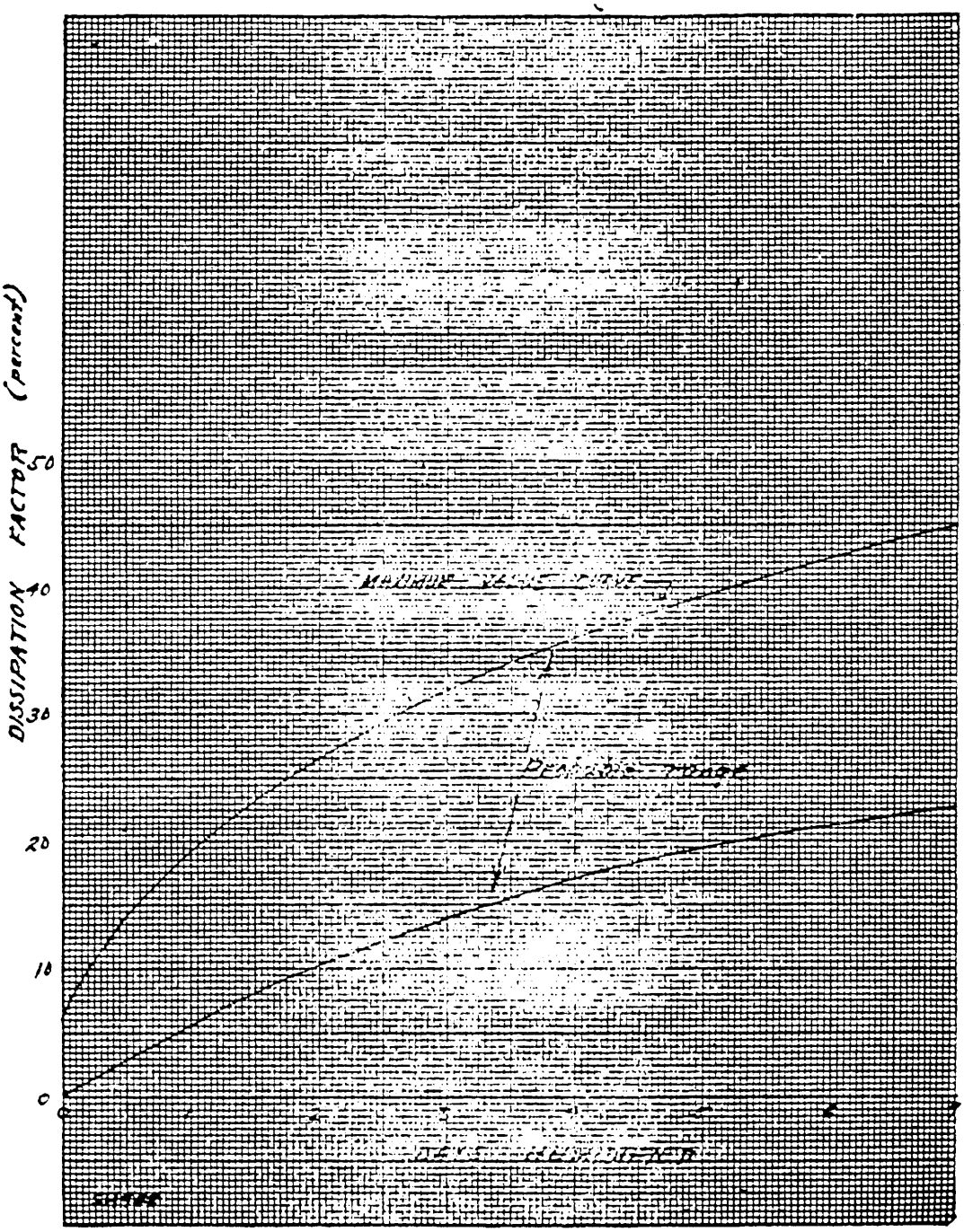


Figure 2 - Dissipation factor.

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INDEX

	Paragraph	Page
A.c. input transients	3.5.2.1.1	4
A.c. voltage	3.5.2.1	4
Airborne noise	3.5.4.1	5
Ambient temperature	3.5.3.2	5
Applicable documents	2.	1
Arrangement of components	3.6.6	7
Attachments, lifting	3.6.3	6
Balances and phases	3.5.2.3	4
Boards, terminal	3.8.10	9
Capacitors	3.8.7	9
Circuit and components protective means	3.5.2.8	4
Classification	1.2	1
Component parts, miscellaneous	3.8.11	10
Components	3.8	8
Components, arrangement of	3.6.6	7
Construction	3.6	6
Contactors and relays	3.8.8	9
Data drawings	3.9.1.1	14
Data, ordering	6.1	22
Definitions	3.4	3
Battery charger	3.4.3	3
Dummy load	3.4.7	3
Full-wave rectification	3.4.4	3
Percent ripple voltage	3.4.6	3
Power supply	3.4.2	3
Rectifying element	3.4.1	3
Ripple voltage	3.4.5	3
Design requirements	3.5	3
Detail requirements	3.1	2
Dielectric strength	3.7.1.2	8
Direct current output	3.5.2.4	4
Draft:		
Forced	3.6.5.2	6
Natural	3.6.5.1	6
Drawings	3.9	10
Data	3.9.1.1	14
Installation	3.9.3	14
Master	3.9.1	10
Preliminary	3.9.2	14
Duty	3.5.2.5	4
Efficiency	3.5.2.9	5
Electrical rating and performance	3.5.2	3
Electromagnetic (radio frequency interference)	3.5.5	6
Enclosure	3.6.1	6
Environmental requirements	3.5.2.9	5
Forced draft	3.6.5.2	6
Frequency	3.5.2.2	4
Fuseholders:		
(With indicators)	3.8.5.1	9
(Without indicators)	3.8.5.2	9
Fuses	3.8.5	9

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Check the source to verify that this is the current version before use.

INDEX (cont'd)

	Paragraph	Page
General examination	4.4	16
Completed unit	4.4.2	23
Major parts	4.4.1	23
General requirements	3.3	3
Humidity	3.5.3.3	5
Indicator lights	3.8.4	8
Inductors, reactors and transformers	3.8.2	8
Input transients, a.c.	3.5.2.1.1	4
Inspection:		
Periodic	4.3.3	19
Routine	4.3.1	18
Selective	4.3.2	18
Installation drawings	3.9.3	14
Insulation	3.7.1	8
Insulation resistance	3.7.1.1	8
Lifting attachments	3.6.3	6
Lights, indicator	3.8.4	8
Load, type of	3.5.2.6	4
Manuals	3.10	14
Master drawings	3.9.1	10
Material requirements	3.7	8
Mechanical	3.5.3.1	5
Meters and meter accessories	3.8.3	8
Miscellaneous component parts	3.8.11	10
Mounting	3.6.2	6
Natural draft	3.6.5.1	6
Noise:		
Airborne	3.5.4.1	5
Structureborne	3.5.4.2	5
Notes	6.	22
Operation	3.5.1	3
Ordering data	6.1	22
Other publications	2.2	2
Output, d.c.	3.5.2.4	4
Painting	3.7.2	8
Parts, repair	3.11	15
Performance and rating, electrical	3.5.2	3
Periodic inspection	4.3.3	19
Periodic, routine and selective inspection	4.3	18
Phases and balances	3.5.2.3	4
Preliminary drawings	3.9.2	14
Preparation for delivery	5.	27
Prohibited components	3.8.12	10
Prohibited materials	3.8.13	10
Qualification	3.2 and 4.2	3, 17
Quality assurance provisions	4.	16
Rating and performance, electrical	3.5.2	3
Reactor, transformer and inductors	3.8.2	8
Relative humidity	3.5.3.3	

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Check the source to verify that this is the current version before use.

INDEX (cont'd)

	Paragraph	Page
Relays and contactors	3.8.8	9
Remote sensing	3.5.2.4.1	4
Repair parts	3.11	15
Repair parts furnished on consolidated shipworth basis	3.11.2	16
Requirements:	3.	2
Construction	3.6	6
Design	3.5	3
Detail	3.1	2
Environmental	3.5.3	5
General	3.3	3
Material	3.7	8
Submarine	3.6.4.1	6
Resistors and rheostats	3.8.6	9
Routine inspection	4.3.1	18
Routine, selective and periodic inspection	4.3	18
Scope	1.1	1
Selective inspection	4.3.2	18
Selective, routine and periodic inspection	4.3	18
Semiconductor devices	3.8.1	8
Size and weight	3.6.4	6
Structureborne noise	3.5.4.2	5
Submarine requirements	3.6.4.1	6
Switches	3.8.9	9
Temperature, ambient	3.5.3.2	5
Terminal boards	3.8.10	9
Tests	4.5	20
Airborne and structureborne noise	4.5.14	27
Creepage and clearance distance	4.5.11	25
Dielectric	4.5.12	25
Effectiveness of enclosure	4.5.3	21
Efficiency	4.5.1	20
Environmental	4.5.13	26
Heating	4.5.4	21
Insulation resistance	4.5.2	21
Performance:	4.5.10	23
Electrical test instruments	4.5.10.7	25
Line voltage transmission	4.5.10.4	24
Operational modes other than voltage regulation	4.5.10.5	24
Resolution	4.5.10.3	24
Ripple voltage	4.5.10.6	25
Test conditions	4.5.10.1	23
Voltage regulation	4.5.10.2	23
Phase balance	4.5.9	23
Shock	4.5.6	21
Starting	4.5.8	23
Vibration	4.5.7	22
Weight	4.5.5	21
Transformers, reactors and inductors	3.8.2	8
Type of load	3.5.2.6	4
Use of resilient mounting of power supply components	3.5.4.3	5

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INDEX (cont'd)

	Paragraph	Page
Ventilation	3.6.5	6
Voltage, a.c.	3.5.2.1	3
Voltage ripple	3.5.2.7	4
Weight and size	3.6.4	6

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